

DATA SCIENCE

Optimizing Customer Retention

with Machine Learning

Proactive identification of customer churn
for strategic decisions & budget optimization.

Data Analyst · ML Specialist

97%

MODEL ACCURACY

Random Forest Classifier

100%

Precision

93%

Recall

0.97

F1-Score

The Challenge: Why Are Customers Leaving?



Revenue Impact

Undetected churn erodes monthly recurring revenue and threatens sustainable growth targets.



Strategic Objective

Beyond prediction — identify which variables drive churn to optimize marketing spend effectively.



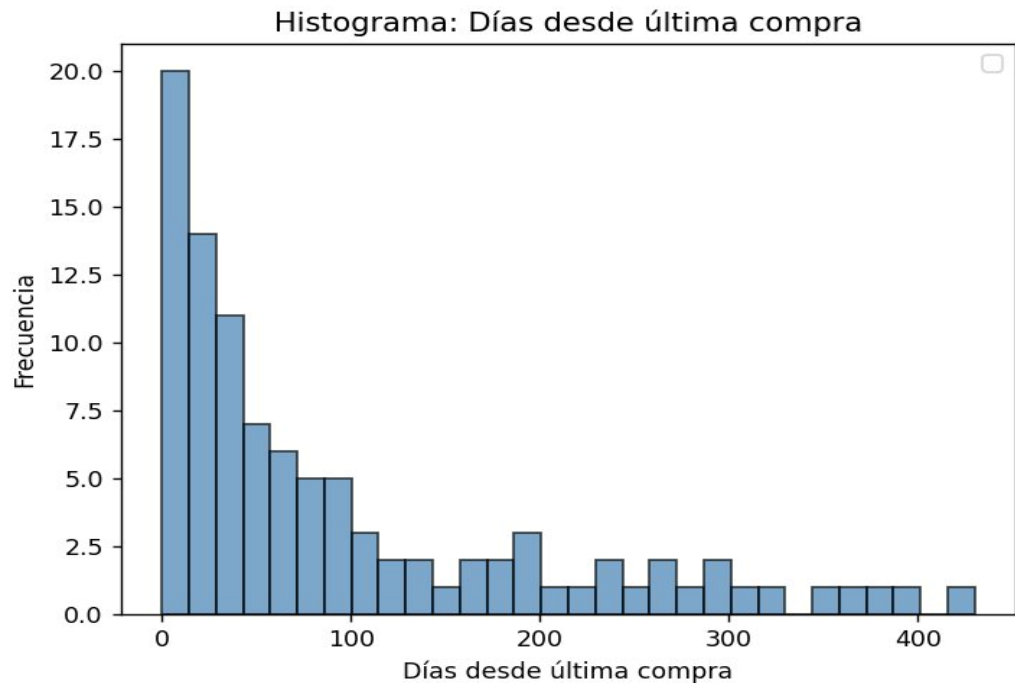
Primary KPI

Reduce the customer churn rate through proactive, model-driven early intervention strategies.



Retaining an existing customer costs 5× less than acquiring a new one — every churn signal matters.

Listening to the Data: Days Since Last Purchase



>30d

Critical inactivity threshold — risk escalates sharply beyond this point

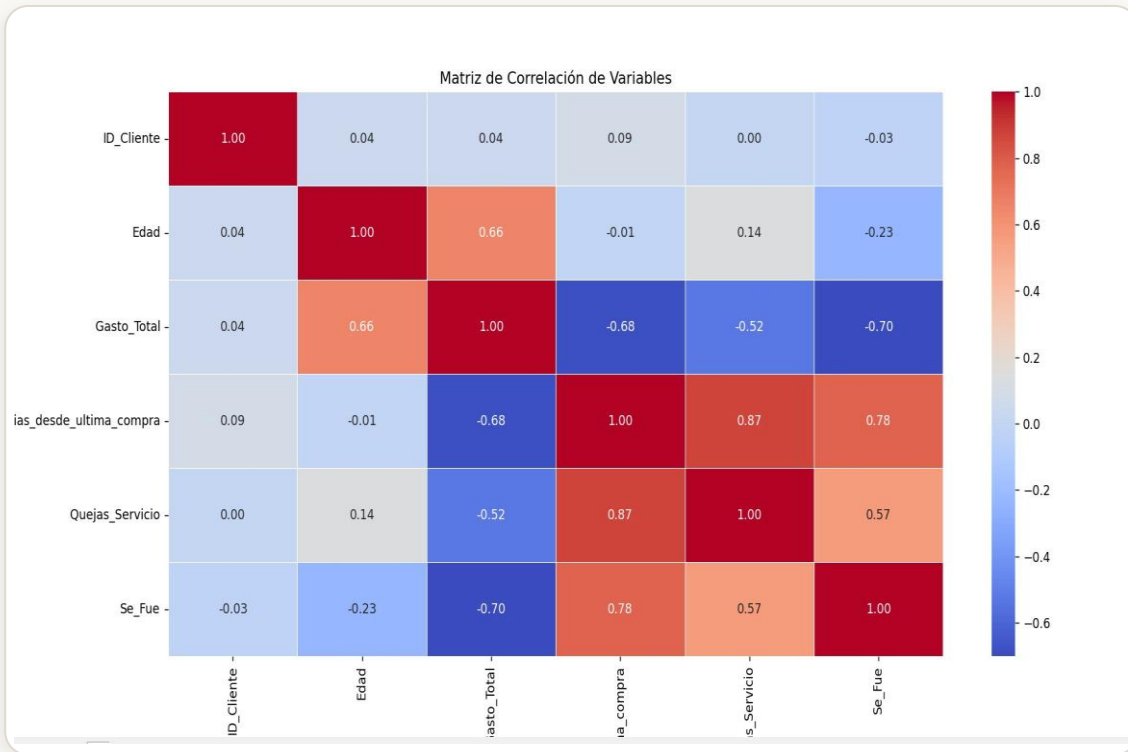
~50%

Of customers purchased within the last 30 days — active core is healthy

Skew

Distribution signals a dangerous long tail of disengaged customers

Connecting the Dots: Variable Relationships



Critical Correlations

+0.78

Inactivity ↔ Churn

Strong positive link — longer gaps dramatically increase churn probability.

-0.70

Spending ↔ Churn

High-value customers are more loyal. VIP clients rarely leave.

+0.87

Complaints ↔ Inactivity

Poor service triggers immediate disengagement — act within 24h.

The Brain: Random Forest Classifier

How It Works

Random Forest is an ensemble of decision trees, each trained on different data subsets. It handles **non-linear relationships** and complex patterns naturally — ideal for real-world churn behavior.

- ✓ Handles imbalanced classes
- ✓ Resistant to overfitting
- ✓ Built-in feature importance ranking

97%

OVERALL ACCURACY

30 customers tested · 28 correctly classified

100%

Precision

Class 1 (Churners)

Zero false alarms

93%

Recall

Class 1 (Churners)

9 of 10 detected

0.97

F1-Score

Macro Average

Perfect balance

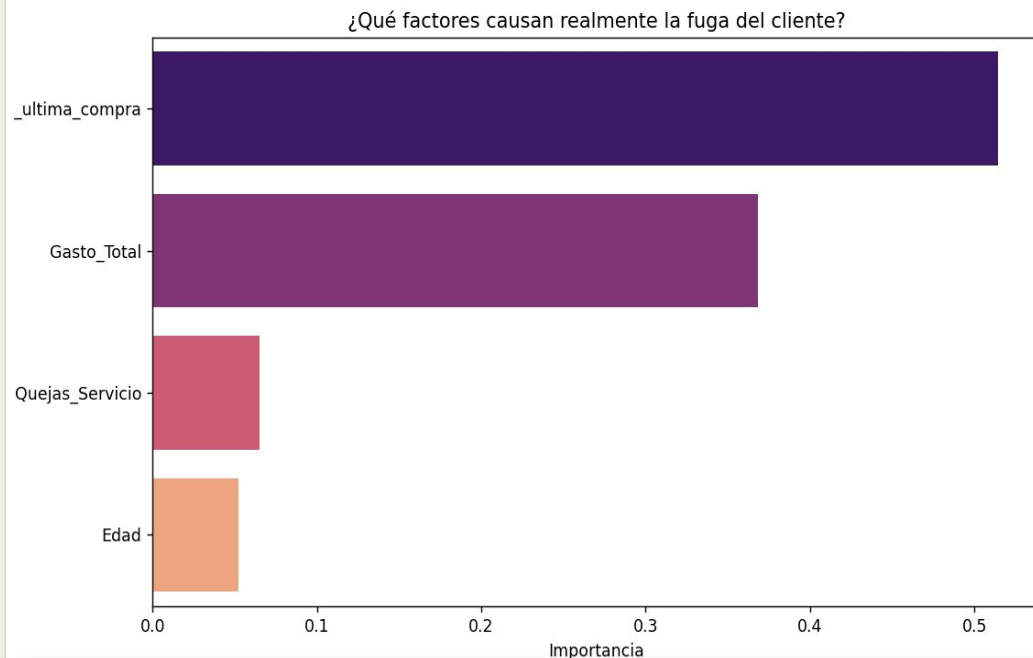
1.00

Recall

Class 0 (Retained)

Zero missed loyal

What Really Drives Customer Churn?



 **Inactivity + Spending together explain 88% of all churn — focus resources here.**

Model Interpretation

51%

Days Since Purchase

Dominant predictor — inactivity alone drives half of all churn

37%

Total Spending

High spenders are significantly more loyal — VIP retention works

7%

Service Complaints

Complaints act as catalysts — they trigger disengagement immediately

5%

Age

Irrelevant — age does not predict churn, simplifying targeting

From Insight to Action

01 Reactivation Campaigns

📄 *Trigger at Day 25 of Inactivity*

Automated personalized outreach before the critical 30-day threshold. Email + push notification + tailored discount offer to recapture at-risk customers.

02 VIP Rescue Plan

📄 *High-Value Customers with Complaints*

Priority support queue for high-spend clients reporting issues. Proactive outreach within 24h of a complaint — preventing the disengagement cascade.

03 Live Risk Dashboard

📄 *Power BI + Model Integration*

Deploy the trained model as a daily scoring pipeline feeding a Power BI dashboard, enabling real-time monitoring of individual customer churn risk scores.

💡 **93% Recall = We can identify and save 9 out of every 10 customers who were about to leave.**